

39. How many different elements does A^n have when A has m elements and n is a positive integer?
40. Show that $A \times B \neq B \times A$, when A and B are nonempty, unless $A = B$.
41. Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.
42. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.
43. Prove or disprove that if A and B are sets, then $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$.
44. Prove or disprove that if A , B , and C are nonempty sets and $A \times B = A \times C$, then $B = C$.
45. Translate each of these quantifications into English and determine its truth value.
- a) $\forall x \in \mathbf{R} (x^2 \neq -1)$ b) $\exists x \in \mathbf{Z} (x^2 = 2)$
 c) $\forall x \in \mathbf{Z} (x^2 > 0)$ d) $\exists x \in \mathbf{R} (x^2 = x)$
46. Translate each of these quantifications into English and determine its truth value.
- a) $\exists x \in \mathbf{R} (x^3 = -1)$ b) $\exists x \in \mathbf{Z} (x + 1 > x)$
 c) $\forall x \in \mathbf{Z} (x - 1 \in \mathbf{Z})$ d) $\forall x \in \mathbf{Z} (x^2 \in \mathbf{Z})$
47. Find the truth set of each of these predicates where the domain is the set of integers.
- a) $P(x): x^2 < 3$ b) $Q(x): x^2 > x$
 c) $R(x): 2x + 1 = 0$
48. Find the truth set of each of these predicates where the domain is the set of integers.
- a) $P(x): x^3 \geq 1$ b) $Q(x): x^2 = 2$
 c) $R(x): x < x^2$
- *49. The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair (a, b) to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$. [Hint: First show that $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$.]
- *50. This exercise presents **Russell's paradox**. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.
- a) Show the assumption that S is a member of S leads to a contradiction.
 b) Show the assumption that S is not a member of S leads to a contradiction.
- By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.
- *51. Describe a procedure for listing all the subsets of a finite set.

2.2 Set Operations

2.2.1 Introduction



Two, or more, sets can be combined in many different ways. For instance, starting with the set of mathematics majors at your school and the set of computer science majors at your school, we can form the set of students who are mathematics majors or computer science majors, the set of students who are joint majors in mathematics and computer science, the set of all students not majoring in mathematics, and so on.

Definition 1

Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

The Venn diagram shown in Figure 1 represents the union of two sets A and B . The area that represents $A \cup B$ is the shaded area within either the circle representing A or the circle representing B .

We will give some examples of the union of sets.

EXAMPLE 1 The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$. ◀

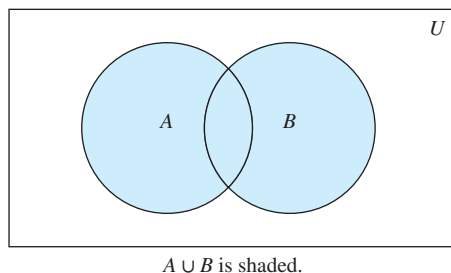


FIGURE 1 Venn diagram of the union of A and B .

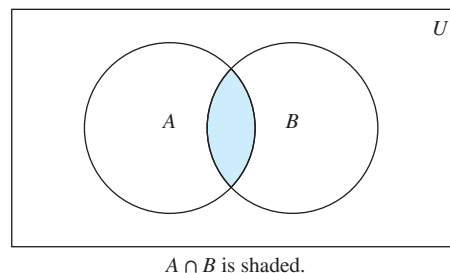


FIGURE 2 Venn diagram of the intersection of A and B .

EXAMPLE 2 The union of the set of all computer science majors at your school and the set of all mathematics majors at your school is the set of students at your school who are majoring either in mathematics or in computer science (or in both). ◀

Definition 2 Let A and B be sets. The *intersection* of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B . This tells us that

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

The Venn diagram shown in Figure 2 represents the intersection of two sets A and B . The shaded area that is within both the circles representing the sets A and B is the area that represents the intersection of A and B .

We give some examples of the intersection of sets.

EXAMPLE 3 The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 3\}$; that is, $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$. ◀

EXAMPLE 4 The intersection of the set of all computer science majors at your school and the set of all mathematics majors is the set of all students who are joint majors in mathematics and computer science. ◀

Definition 3 Two sets are called *disjoint* if their intersection is the empty set.

EXAMPLE 5 Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Because $A \cap B = \emptyset$, A and B are disjoint. ◀

Be careful not to overcount!

We are often interested in finding the cardinality of a union of two finite sets A and B . Note that $|A| + |B|$ counts each element that is in A but not in B or in B but not in A exactly once, and each element that is in both A and B exactly twice. Thus, if the number of elements that are in both A and B is subtracted from $|A| + |B|$, elements in $A \cap B$ will be counted only once. Hence,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The generalization of this result to unions of an arbitrary number of sets is called the **principle of inclusion–exclusion**. The principle of inclusion–exclusion is an important technique used in enumeration. We will discuss this principle and other counting techniques in detail in Chapters 6 and 8.

There are other important ways to combine sets.

Definition 4

Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the *complement of B with respect to A* .

Remark: The difference of sets A and B is sometimes denoted by $A \setminus B$.

An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that

$$A - B = \{x \mid x \in A \wedge x \notin B\}.$$

The Venn diagram shown in Figure 3 represents the difference of the sets A and B . The shaded area inside the circle that represents A and outside the circle that represents B is the area that represents $A - B$.

We give some examples of differences of sets.

EXAMPLE 6

The difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{5\}$; that is, $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$. This is different from the difference of $\{1, 2, 3\}$ and $\{1, 3, 5\}$, which is the set $\{2\}$. 

EXAMPLE 7

The difference of the set of computer science majors at your school and the set of mathematics majors at your school is the set of all computer science majors at your school who are not also mathematics majors. 

Once the universal set U has been specified, the **complement** of a set can be defined.

Definition 5

Let U be the universal set. The *complement* of the set A , denoted by $\bar{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.

Remark: The definition of the complement of A depends on a particular universal set U . This definition makes sense for any superset U of A . If we want to identify the universal set U , we would write “the complement of A with respect to the set U .”

An element belongs to $\bar{A}$ if and only if $x \notin A$. This tells us that

$$\bar{A} = \{x \in U \mid x \notin A\}.$$

In Figure 4 the shaded area outside the circle representing A is the area representing $\bar{A}$.

We give some examples of the complement of a set.

EXAMPLE 8

Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then $\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$. 

EXAMPLE 9

Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Then $\bar{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. 

It is left to the reader (Exercise 21) to show that we can express the difference of A and B as the intersection of A and the complement of B . That is,

$$A - B = A \cap \bar{B}.$$

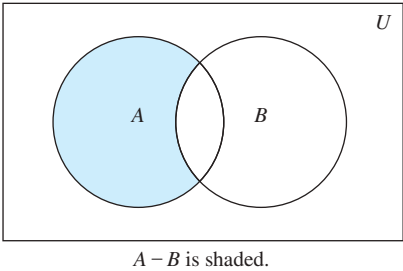


FIGURE 3 Venn diagram for the difference of A and B .

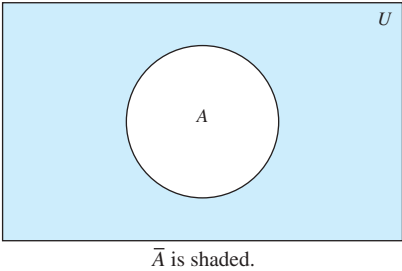


FIGURE 4 Venn diagram for the complement of the set A .

2.2.2 Set Identities

Set identities and propositional equivalences are just special cases of identities for Boolean algebra.

Table 1 lists the most important identities of unions, intersections, and complements of sets. We will prove several of these identities here, using three different methods. These methods are presented to illustrate that there are often many different approaches to the solution of a problem. The proofs of the remaining identities will be left as exercises. The reader should note the similarity between these set identities and the logical equivalences discussed in Section 1.3. (Compare Table 6 of Section 1.6 and Table 1.) In fact, the set identities given can be proved directly from the corresponding logical equivalences. Furthermore, both are special cases of identities that hold for Boolean algebra (discussed in Chapter 12).

TABLE 1 Set Identities.	
Identity	Name
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\bar{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \bar{A} \cup \bar{B}$ $\overline{A \cup B} = \bar{A} \cap \bar{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

Before we discuss different approaches for proving set identities, we briefly discuss the role of Venn diagrams. Although these diagrams can help us understand sets constructed using two or three **atomic sets** (the sets used to construct more complicated combinations of these sets), they provide far less insight when four or more atomic sets are involved. Venn diagrams for four or more sets are quite complex because it is necessary to use ellipses rather than circles to represent the sets. This is necessary to ensure that every possible combination of the sets is represented by a nonempty region. Although Venn diagrams can provide an informal proof for some identities, such proofs should be formalized using one of the three methods we will now describe.

This identity says that the complement of the intersection of two sets is the union of their complements.

One way to show that two sets are equal is to show that each is a subset of the other. Recall that to show that one set is a subset of a second set, we can show that if an element belongs to the first set, then it must also belong to the second set. We generally use a direct proof to do this. We illustrate this type of proof by establishing the first of De Morgan's laws.

EXAMPLE 10 Prove that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Extra Examples ➤

Solution: We will prove that the two sets $\overline{A \cap B}$ and $\bar{A} \cup \bar{B}$ are equal by showing that each set is a subset of the other.

First, we will show that $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$. We do this by showing that if x is in $\overline{A \cap B}$, then it must also be in $\bar{A} \cup \bar{B}$. Now suppose that $x \in \overline{A \cap B}$. By the definition of complement, $x \notin A \cap B$. Using the definition of intersection, we see that the proposition $\neg((x \in A) \wedge (x \in B))$ is true.

By applying De Morgan's law for propositions, we see that $\neg(x \in A) \vee \neg(x \in B)$. Using the definition of negation of propositions, we have $x \notin A$ or $x \notin B$. Using the definition of the complement of a set, we see that this implies that $x \in \bar{A}$ or $x \in \bar{B}$. Consequently, by the definition of union, we see that $x \in \bar{A} \cup \bar{B}$. We have now shown that $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$.

Next, we will show that $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$. We do this by showing that if x is in $\bar{A} \cup \bar{B}$, then it must also be in $\overline{A \cap B}$. Now suppose that $x \in \bar{A} \cup \bar{B}$. By the definition of union, we know that $x \in \bar{A}$ or $x \in \bar{B}$. Using the definition of complement, we see that $x \notin A$ or $x \notin B$. Consequently, the proposition $\neg(x \in A) \vee \neg(x \in B)$ is true.

By De Morgan's law for propositions, we conclude that $\neg((x \in A) \wedge (x \in B))$ is true. By the definition of intersection, it follows that $\neg(x \in A \cap B)$. We now use the definition of complement to conclude that $x \in \overline{A \cap B}$. This shows that $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$.

Because we have shown that each set is a subset of the other, the two sets are equal, and the identity is proved. ◀

We can more succinctly express the reasoning used in Example 10 using set builder notation, as Example 11 illustrates.

EXAMPLE 11 Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
 \overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{by definition of complement} \\
 &= \{x \mid \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol} \\
 &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
 &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan law for logical equivalences} \\
 &= \{x \mid x \notin A \vee x \notin B\} && \text{by definition of does not belong symbol} \\
 &= \{x \mid x \in \bar{A} \vee x \in \bar{B}\} && \text{by definition of complement} \\
 &= \{x \mid x \in \bar{A} \cup \bar{B}\} && \text{by definition of union} \\
 &= \bar{A} \cup \bar{B} && \text{by meaning of set builder notation}
 \end{aligned}$$

Note that besides the definitions of complement, union, set membership, and set builder notation, this proof uses the second De Morgan law for logical equivalences. ◀

Proving a set identity involving more than two sets by showing each side of the identity is a subset of the other often requires that we keep track of different cases, as illustrated by the proof in Example 12 of one of the distributive laws for sets.

EXAMPLE 12 Prove the second distributive law from Table 1, which states that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ for all sets A , B , and C .

Solution: We will prove this identity by showing that each side is a subset of the other side.

Suppose that $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. By the definition of union, it follows that $x \in A$, and $x \in B$ or $x \in C$ (or both). In other words, we know that the compound proposition $(x \in A) \wedge ((x \in B) \vee (x \in C))$ is true. By the distributive law for conjunction over disjunction, it follows that $((x \in A) \wedge (x \in B)) \vee ((x \in A) \wedge (x \in C))$. We conclude that either $x \in A$ and $x \in B$, or $x \in A$ and $x \in C$. By the definition of intersection, it follows that $x \in A \cap B$ or $x \in A \cap C$. Using the definition of union, we conclude that $x \in (A \cap B) \cup (A \cap C)$. We conclude that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Now suppose that $x \in (A \cap B) \cup (A \cap C)$. Then, by the definition of union, $x \in A \cap B$ or $x \in A \cap C$. By the definition of intersection, it follows that $x \in A$ and $x \in B$ or that $x \in A$ and $x \in C$. From this we see that $x \in A$, and $x \in B$ or $x \in C$. Consequently, by the definition of union we see that $x \in A$ and $x \in B \cup C$. Furthermore, by the definition of intersection, it follows that $x \in A \cap (B \cup C)$. We conclude that $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. This completes the proof of the identity. ◀

Set identities can also be proved using **membership tables**. We consider each combination of the atomic sets (that is, the original sets used to produce the sets on each side) that an element can belong to and verify that elements in the same combinations of sets belong to both the sets in the identity. To indicate that an element is in a set, a 1 is used; to indicate that an element is not in a set, a 0 is used. (The reader should note the similarity between membership tables and truth tables.)

EXAMPLE 13 Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution: The membership table for these combinations of sets is shown in Table 2. This table has eight rows. Because the columns for $A \cap (B \cup C)$ and $(A \cap B) \cup (A \cap C)$ are the same, the identity is valid. ◀

TABLE 2 A Membership Table for the Distributive Property.							
A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Once we have proved set identities, we can use them to prove new identities. In particular, we can apply a string of identities, one in each step, to take us from one side of a desired identity to the other. It is helpful to explicitly state the identity that is used in each step, as we do in Example 14.

EXAMPLE 14 Let A , B , and C be sets. Show that

$$\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}.$$

Solution: We have

$$\begin{aligned} \overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{(B \cap C)} && \text{by the first De Morgan law} \\ &= \overline{A} \cap (\overline{B} \cup \overline{C}) && \text{by the second De Morgan law} \\ &= (\overline{B} \cup \overline{C}) \cap \overline{A} && \text{by the commutative law for intersections} \\ &= (\overline{C} \cup \overline{B}) \cap \overline{A} && \text{by the commutative law for unions.} \end{aligned}$$

We summarize the three different ways for proving set identities in Table 3.

TABLE 3 Methods of Proving Set Identities.	
Description	Method
Subset method	Show that each side of the identity is a subset of the other side.
Membership table	For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side
Apply existing identities	Start with one side, transform it into the other side using a sequence of steps by applying an established identity.

2.2.3 Generalized Unions and Intersections

Because unions and intersections of sets satisfy associative laws, the sets $A \cup B \cup C$ and $A \cap B \cap C$ are well defined; that is, the meaning of this notation is unambiguous when A , B , and C are sets. That is, we do not have to use parentheses to indicate which operation comes first because $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$. Note that $A \cup B \cup C$ contains those elements that are in at least one of the sets A , B , and C , and that $A \cap B \cap C$ contains those elements that are in all of A , B , and C . These combinations of the three sets, A , B , and C , are shown in Figure 5.

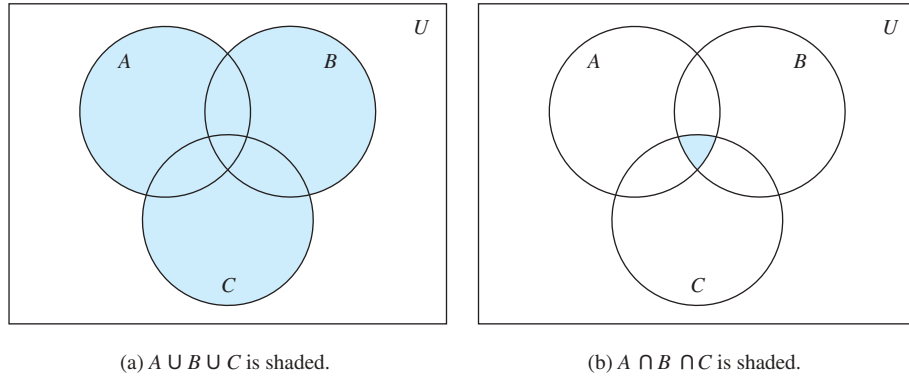
EXAMPLE 15 Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

Solution: The set $A \cup B \cup C$ contains those elements in at least one of A , B , and C . Hence,

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}.$$

The set $A \cap B \cap C$ contains those elements in all three of A , B , and C . Thus,

$$A \cap B \cap C = \{0\}.$$

**FIGURE 5** The union and intersection of A , B , and C .

We can also consider unions and intersections of an arbitrary number of sets. We introduce these definitions.

Definition 6

The *union* of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

We use the notation

$$A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^n A_i$$

to denote the union of the sets $A_1, A_2, \dots, A_n$.

Definition 7

The *intersection* of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

We use the notation

$$A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^n A_i$$

to denote the intersection of the sets $A_1, A_2, \dots, A_n$. We illustrate generalized unions and intersections with Example 16.

EXAMPLE 16 For $i = 1, 2, \dots$, let $A_i = \{i, i + 1, i + 2, \dots\}$. Then,

Extra Examples ➤

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{1, 2, 3, \dots\},$$

and

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{n, n + 1, n + 2, \dots\} = A_n.$$

We can extend the notation we have introduced for unions and intersections to other families of sets. In particular, to denote the union of the infinite family of sets $A_1, A_2, \dots, A_n, \dots$, we use the notation

$$A_1 \cup A_2 \cup \dots \cup A_n \cup \dots = \bigcup_{i=1}^{\infty} A_i.$$

Similarly, the intersection of these sets is denoted by

$$A_1 \cap A_2 \cap \dots \cap A_n \cap \dots = \bigcap_{i=1}^{\infty} A_i.$$

More generally, when I is a set, the notations $\bigcap_{i \in I} A_i$ and $\bigcup_{i \in I} A_i$ are used to denote the intersection and union of the sets A_i for $i \in I$, respectively. Note that we have $\bigcap_{i \in I} A_i = \{x \mid \forall i \in I (x \in A_i)\}$ and $\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \in A_i)\}$.

EXAMPLE 17 Suppose that $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Then,

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \{1, 2, 3, \dots, i\} = \{1, 2, 3, \dots\} = \mathbf{Z}^+$$

and

$$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} \{1, 2, 3, \dots, i\} = \{1\}.$$

To see that the union of these sets is the set of positive integers, note that every positive integer n is in at least one of the sets, because it belongs to $A_n = \{1, 2, \dots, n\}$, and every element of the sets in the union is a positive integer. To see that the intersection of these sets is the set $\{1\}$, note that the only element that belongs to all the sets $A_1, A_2, \dots$ is 1. To see this note that $A_1 = \{1\}$ and $1 \in A_i$ for $i = 1, 2, \dots$. 

2.2.4 Computer Representation of Sets

There are various ways to represent sets using a computer. One method is to store the elements of the set in an unordered fashion. However, if this is done, the operations of computing the union, intersection, or difference of two sets would be time consuming, because each of these operations would require a large amount of searching for elements. We will present a method for storing elements using an arbitrary ordering of the elements of the universal set. This method of representing sets makes computing combinations of sets easy.

Assume that the universal set U is finite (and of reasonable size so that the number of elements of U is not larger than the memory size of the computer being used). First, specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$. Represent a subset A of U with the bit string of length n , where the i th bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A . Example 18 illustrates this technique.

EXAMPLE 18 Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order; that is, $a_i = i$. What bit strings represent the subset of all odd integers in U , the subset of all even integers in U , and the subset of integers not exceeding 5 in U ?

Solution: The bit string that represents the set of odd integers in U , namely, $\{1, 3, 5, 7, 9\}$, has a one bit in the first, third, fifth, seventh, and ninth positions, and a zero elsewhere. It is

10 1010 1010.

(We have split this bit string of length ten into blocks of length four for easy reading.) Similarly, we represent the subset of all even integers in U , namely, $\{2, 4, 6, 8, 10\}$, by the string

01 0101 0101.

The set of all integers in U that do not exceed 5, namely, $\{1, 2, 3, 4, 5\}$, is represented by the string

11 1110 0000. 

Using bit strings to represent sets, it is easy to find complements of sets and unions, intersections, and differences of sets. To find the bit string for the complement of a set from the bit string for that set, we simply change each 1 to a 0 and each 0 to 1, because $x \in A$ if and only if $x \notin \bar{A}$. Note that this operation corresponds to taking the negation of each bit when we associate a bit with a truth value—with 1 representing true and 0 representing false.

EXAMPLE 19 We have seen that the bit string for the set $\{1, 3, 5, 7, 9\}$ (with universal set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$) is

10 1010 1010.

What is the bit string for the complement of this set?

Solution: The bit string for the complement of this set is obtained by replacing 0s with 1s and vice versa. This yields the string

01 0101 0101,

which corresponds to the set $\{2, 4, 6, 8, 10\}$. 

To obtain the bit string for the union and intersection of two sets we perform bitwise Boolean operations on the bit strings representing the two sets. The bit in the i th position of the bit string of the union is 1 if either of the bits in the i th position in the two strings is 1 (or both are 1), and is 0 when both bits are 0. Hence, the bit string for the union is the bitwise *OR* of the bit strings for the two sets. The bit in the i th position of the bit string of the intersection is 1 when the bits in the corresponding position in the two strings are both 1, and is 0 when either of the two bits is 0 (or both are). Hence, the bit string for the intersection is the bitwise *AND* of the bit strings for the two sets.

EXAMPLE 20 The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

Solution: The bit string for the union of these sets is

$11\ 1110\ 0000 \vee 10\ 1010\ 1010 = 11\ 1110\ 1010$,

which corresponds to the set $\{1, 2, 3, 4, 5, 7, 9\}$. The bit string for the intersection of these sets is

$11\ 1110\ 0000 \wedge 10\ 1010\ 1010 = 10\ 1010\ 0000$,

which corresponds to the set $\{1, 3, 5\}$. 

2.2.5 Multisets

Sometimes the number of times that an element occurs in an unordered collection matters. A **multiset** (short for multiple-membership set) is an unordered collection of elements where an element can occur as a member more than once. We can use the same notation for a multiset as we do for a set, but each element is listed the number of times it occurs. (Recall that in a set, an element either belongs to a set or it does not. Listing it more than once does not affect the membership of this element in the set.) So, the multiset denoted by $\{a, a, a, b, b\}$ is the multiset that contains the element a thrice and the element b twice. When we use this notation, it must be clear that we are working with multisets and not ordinary sets. We can avoid this ambiguity by using an alternate notation for multisets. The notation $\{m_1 \cdot a_1, m_2 \cdot a_2, \dots, m_r \cdot a_r\}$ denotes the multiset with element a_1 occurring m_1 times, element a_2 occurring m_2 times, and so on. The numbers m_i , $i = 1, 2, \dots, r$, are called the **multiplicities** of the elements a_i , $i = 1, 2, \dots, r$. (Elements not in a multiset are assigned 0 as their multiplicity in this set.) The cardinality of a multiset is defined to be the sum of the multiplicities of its elements. The word *multiset* was introduced by Nicolaas Govert de Bruijn in the 1970s, but the concept dates back to the 12th century work of the Indian mathematician Bhaskaracharya.

Let P and Q be multisets. The **union** of the multisets P and Q is the multiset in which the multiplicity of an element is the maximum of its multiplicities in P and Q . The **intersection** of P and Q is the multiset in which the multiplicity of an element is the minimum of its multiplicities in P and Q . The **difference** of P and Q is the multiset in which the multiplicity of an element is the multiplicity of the element in P less its multiplicity in Q unless this difference is negative, in which case the multiplicity is 0. The **sum** of P and Q is the multiset in which the multiplicity of an element is the sum of multiplicities in P and Q . The union, intersection, and difference of P and Q are denoted by $P \cup Q$, $P \cap Q$, and $P - Q$, respectively (where these operations should not be confused with the analogous operations for sets). The sum of P and Q is denoted by $P + Q$.

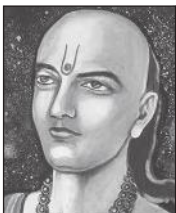
EXAMPLE 21 Suppose that P and Q are the multisets $\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$ and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$, respectively. Find $P \cup Q$, $P \cap Q$, $P - Q$, and $P + Q$.

Solution: We have

$$\begin{aligned} P \cup Q &= \{\max(4, 3) \cdot a, \max(1, 4) \cdot b, \max(3, 0) \cdot c, \max(0, 2) \cdot d\} \\ &= \{4 \cdot a, 4 \cdot b, 3 \cdot c, 2 \cdot d\}, \end{aligned}$$

$$\begin{aligned} P \cap Q &= \{\min(4, 3) \cdot a, \min(1, 4) \cdot b, \min(3, 0) \cdot c, \min(0, 2) \cdot d\} \\ &= \{3 \cdot a, 1 \cdot b, 0 \cdot c, 0 \cdot d\} = \{3 \cdot a, 1 \cdot b\}, \end{aligned}$$

Links



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BHASKARACHARYA (1114–1185) Bhaskaracharya was born in Bijapur in the Indian state of Karnataka. (Bhaskaracharya's name was actually Bhaskara, but the title Acharya, which means teacher, was added honorifically.) His father was a well-known scholar and a famous astrologer. Bhaskaracharya was head of the astronomical observatory at Ujjain, the leading Indian mathematical center of the day. He is considered to be the greatest mathematician of medieval India. Bhaskaracharya made discoveries in many parts of mathematics, including geometry, plane and spherical trigonometry, algebra, number theory, and combinatorics. Bhaskaracharya described the principles of differential calculus, which he applied to astronomical problems, predating the works of Newton and Leibniz by more than 500 years. In number theory he made many discoveries about Diophantine equations, the study of the solution in integers of equations, which were rediscovered more than 600 years later. His greatest work is the *Crown of Treatises (Siddhanta Shiromani)*, which includes four main parts, covering arithmetic, algebra, mathematics of the planets, and spheres.

$$\begin{aligned}
 P - Q &= \{\max(4 - 3, 0) \cdot a, \max(1 - 4, 0) \cdot b, \max(3 - 0, 0) \cdot c, \max(0 - 2, 0) \cdot d\} \\
 &= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\}, \text{ and}
 \end{aligned}$$

$$\begin{aligned}
 P + Q &= \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} \\
 &= \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\}.
 \end{aligned}$$

Exercises

- Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.
 - $A \cap B$
 - $A \cup B$
 - $A - B$
 - $B - A$
 - Suppose that A is the set of sophomores at your school and B is the set of students in discrete mathematics at your school. Express each of these sets in terms of A and B .
 - the set of sophomores taking discrete mathematics in your school
 - the set of sophomores at your school who are not taking discrete mathematics
 - the set of students at your school who either are sophomores or are taking discrete mathematics
 - the set of students at your school who either are not sophomores or are not taking discrete mathematics
 - Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find
 - $A \cup B$.
 - $A \cap B$.
 - $A - B$.
 - $B - A$.
 - Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find
 - $A \cup B$.
 - $A \cap B$.
 - $A - B$.
 - $B - A$.
- In Exercises 5–10 assume that A is a subset of some underlying universal set U .
- Prove the complementation law in Table 1 by showing that $\overline{\overline{A}} = A$.
 - Prove the identity laws in Table 1 by showing that
 - $A \cup \emptyset = A$.
 - $A \cap U = A$.
 - Prove the domination laws in Table 1 by showing that
 - $A \cup U = U$.
 - $A \cap \emptyset = \emptyset$.
 - Prove the idempotent laws in Table 1 by showing that
 - $A \cup A = A$.
 - $A \cap A = A$.
 - Prove the complement laws in Table 1 by showing that
 - $A \cup \overline{A} = U$.
 - $A \cap \overline{A} = \emptyset$.
 - Show that
 - $A - \emptyset = A$.
 - $\emptyset - A = \emptyset$.
 - Let A and B be sets. Prove the commutative laws from Table 1 by showing that
 - $A \cup B = B \cup A$.
 - $A \cap B = B \cap A$.
 - Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.
 - Prove the second absorption law from Table 1 by showing that if A and B are sets, then $A \cap (A \cup B) = A$.
 - Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$, and $A \cap B = \{3, 6, 9\}$.
 - Prove the second De Morgan law in Table 1 by showing that if A and B are sets, then $\overline{A \cup B} = \overline{A} \cap \overline{B}$
 - by showing each side is a subset of the other side.
 - using a membership table.
 - Let A and B be sets. Show that
 - $(A \cap B) \subseteq A$.
 - $A \subseteq (A \cup B)$.
 - $A - B \subseteq A$.
 - $A \cap (B - A) = \emptyset$.
 - $A \cup (B - A) = A \cup B$.
 - Show that if A and B are sets in a universe U then $A \subseteq B$ if and only if $\overline{A} \cup B = U$.
 - Given sets A and B in a universe U , draw the Venn diagrams of each of these sets.
 - $A \rightarrow B = \{x \in U \mid x \in A \rightarrow x \in B\}$
 - $A \leftrightarrow B = \{x \in U \mid x \in A \leftrightarrow x \in B\}$
 - Show that if A , B , and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$
 - by showing each side is a subset of the other side.
 - using a membership table.
 - Let A , B , and C be sets. Show that
 - $(A \cup B) \subseteq (A \cup B \cup C)$.
 - $(A \cap B \cap C) \subseteq (A \cap B)$.
 - $(A - B) - C \subseteq A - C$.
 - $(A - C) \cap (C - B) = \emptyset$.
 - $(B - A) \cup (C - A) = (B \cup C) - A$.
 - Show that if A and B are sets, then
 - $A - B = A \cap \overline{B}$.
 - $(A \cap B) \cup (A \cap \overline{B}) = A$.
 - Show that if A and B are sets with $A \subseteq B$, then
 - $A \cup B = B$.
 - $A \cap B = A$.
 - Prove the first associative law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap C$.
 - Prove the second associative law from Table 1 by showing that if A , B , and C are sets, then $A \cap (B \cup C) = (A \cap B) \cup C$.
 - Prove the first distributive law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

26. Let A , B , and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.
27. Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find
- $A \cap B \cap C$.
 - $A \cup B \cup C$.
 - $(A \cup B) \cap C$.
 - $(A \cap B) \cup C$.
28. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .
- $A \cap (B \cup C)$
 - $\overline{A} \cap \overline{B} \cap \overline{C}$
 - $(A - B) \cup (A - C) \cup (B - C)$
29. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .
- $A \cap (B - C)$
 - $(A \cap B) \cup (A \cap C)$
 - $(A \cap \overline{B}) \cup (A \cap \overline{C})$
30. Draw the Venn diagrams for each of these combinations of the sets A , B , C , and D .
- $(A \cap B) \cup (C \cap D)$
 - $\overline{A} \cup \overline{B} \cup \overline{C} \cup \overline{D}$
 - $A - (B \cap C \cap D)$
31. What can you say about the sets A and B if we know that
- $A \cup B = A$?
 - $A \cap B = A$?
 - $A - B = A$?
 - $A \cap B = B \cap A$?
 - $A - B = B - A$?
32. Can you conclude that $A = B$ if A , B , and C are sets such that
- $A \cup C = B \cup C$?
 - $A \cap C = B \cap C$?
 - $A \cup C = B \cup C$ and $A \cap C = B \cap C$?
33. Let A and B be subsets of a universal set U . Show that $A \subseteq B$ if and only if $\overline{B} \subseteq \overline{A}$.
34. Let A , B , and C be sets. Use the identity $A - B = A \cap \overline{B}$, which holds for any sets A and B , and the identities from Table 1 to show that $(A - B) \cap (B - C) \cap (A - C) = \emptyset$.
35. Let A , B , and C be sets. Use the identities in Table 1 to show that $(A \cup B) \cap (B \cup C) \cap (A \cup C) = A \cap B \cap C$.
36. Prove or disprove that for all sets A , B , and C , we have
- $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
 - $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
37. Prove or disprove that for all sets A , B , and C , we have
- $A \times (B - C) = (A \times B) - (A \times C)$.
 - $A \times (B \cup C) = A \times (B \cup C)$.
- The **symmetric difference** of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B .
38. Find the symmetric difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$.
39. Find the symmetric difference of the set of computer science majors at a school and the set of mathematics majors at this school.
40. Draw a Venn diagram for the symmetric difference of the sets A and B .
41. Show that $A \oplus B = (A \cup B) - (A \cap B)$.
42. Show that $A \oplus B = (A - B) \cup (B - A)$.
43. Show that if A is a subset of a universal set U , then
- $A \oplus A = \emptyset$.
 - $A \oplus \emptyset = A$.
 - $A \oplus U = \overline{A}$.
 - $A \oplus \overline{A} = U$.

44. Show that if A and B are sets, then

$$\text{a) } A \oplus B = B \oplus A. \quad \text{b) } (A \oplus B) \oplus B = A.$$

45. What can you say about the sets A and B if $A \oplus B = A$?

*46. Determine whether the symmetric difference is associative; that is, if A , B , and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?

*47. Suppose that A , B , and C are sets such that $A \oplus C = B \oplus C$. Must it be the case that $A = B$?

48. If A , B , C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus C) \oplus (B \oplus D)$?

49. If A , B , C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus D) \oplus (B \oplus C)$?

50. Show that if A and B are finite sets, then $A \cup B$ is a finite set.

51. Show that if A is an infinite set, then whenever B is a set, $A \cup B$ is also an infinite set.

*52. Show that if A , B , and C are sets, then

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

(This is a special case of the inclusion-exclusion principle, which will be studied in Chapter 8.)

53. Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find

$$\text{a) } \bigcup_{i=1}^n A_i. \quad \text{b) } \bigcap_{i=1}^n A_i.$$

54. Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Find

$$\text{a) } \bigcup_{i=1}^n A_i. \quad \text{b) } \bigcap_{i=1}^n A_i.$$

55. Let A_i be the set of all nonempty bit strings (that is, bit strings of length at least one) of length not exceeding i . Find

$$\text{a) } \bigcup_{i=1}^n A_i. \quad \text{b) } \bigcap_{i=1}^n A_i.$$

56. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

- $A_i = \{i, i+1, i+2, \dots\}$.
- $A_i = \{0, i\}$.
- $A_i = (0, i)$, that is, the set of real numbers x with $0 < x < i$.
- $A_i = (i, \infty)$, that is, the set of real numbers x with $x > i$.

57. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

- $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$.
- $A_i = \{-i, i\}$.
- $A_i = [-i, i]$, that is, the set of real numbers x with $-i \leq x \leq i$.
- $A_i = [i, \infty)$, that is, the set of real numbers x with $x \geq i$.

58. Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.

- $\{3, 4, 5\}$
- $\{1, 3, 6, 10\}$
- $\{2, 3, 4, 7, 8, 9\}$

59. Using the same universal set as in the last exercise, find the set specified by each of these bit strings.

- a) 11 1100 1111
- b) 01 0111 1000
- c) 10 0000 0001

60. What subsets of a finite universal set do these bit strings represent?

- a) the string with all zeros
- b) the string with all ones

61. What is the bit string corresponding to the difference of two sets?

62. What is the bit string corresponding to the symmetric difference of two sets?

63. Show how bitwise operations on bit strings can be used to find these combinations of $A = \{a, b, c, d, e\}$, $B = \{b, c, d, g, p, t, v\}$, $C = \{c, e, i, o, u, x, y, z\}$, and $D = \{d, e, h, i, n, o, t, u, x, y\}$.

- a) $A \cup B$
- b) $A \cap B$
- c) $(A \cup D) \cap (B \cup C)$
- d) $A \cup B \cup C \cup D$

64. How can the union and intersection of n sets that all are subsets of the universal set U be found using bit strings?

The **successor** of the set A is the set $A \cup \{A\}$.

65. Find the successors of the following sets.

- a) $\{1, 2, 3\}$
- b) $\emptyset$
- c) $\{\emptyset\}$
- d) $\{\emptyset, \{\emptyset\}\}$

66. How many elements does the successor of a set with n elements have?

67. Let A and B be the multisets $\{3 \cdot a, 2 \cdot b, 1 \cdot c\}$ and $\{2 \cdot a, 3 \cdot b, 4 \cdot d\}$, respectively. Find

- a) $A \cup B$
- b) $A \cap B$
- c) $A - B$
- d) $B - A$
- e) $A + B$

68. Assume that $a \in A$, where A is a set. Which of these statements are true and which are false, where all sets shown are ordinary sets, and not multisets. Explain each answer.

- a) $\{a, a\} \cup \{a, a, a\} = \{a, a, a, a, a\}$
- b) $\{a, a\} \cup \{a, a, a\} = \{a\}$
- c) $\{a, a\} \cap \{a, a, a\} = \{a, a\}$
- d) $\{a, a\} \cap \{a, a, a\} = \{a\}$
- e) $\{a, a, a\} - \{a, a\} = \{a\}$

69. Answer the same questions as posed in Exercise 68 where all sets are multisets, and not ordinary sets.

70. Suppose that A is the multiset that has as its elements the types of computer equipment needed by one department of a university and the multiplicities are the number of pieces of each type needed, and B is the analogous multiset for a second department of the university. For instance, A could be the multiset $\{107 \cdot \text{personal computers}, 44 \cdot \text{routers}, 6 \cdot \text{servers}\}$ and B could be the multiset $\{14 \cdot \text{personal computers}, 6 \cdot \text{routers}, 2 \cdot \text{mainframes}\}$.

- a) What combination of A and B represents the equipment the university should buy assuming both departments use the same equipment?
- b) What combination of A and B represents the equipment that will be used by both departments if both departments use the same equipment?

c) What combination of A and B represents the equipment that the second department uses, but the first department does not, if both departments use the same equipment?

d) What combination of A and B represents the equipment that the university should purchase if the departments do not share equipment?

The **Jaccard similarity** $J(A, B)$ of the finite sets A and B is $J(A, B) = |A \cap B| / |A \cup B|$, with $J(\emptyset, \emptyset) = 1$. The **Jaccard distance** $d_J(A, B)$ between A and B equals $d_J(A, B) = 1 - J(A, B)$.

71. Find $J(A, B)$ and $d_J(A, B)$ for these pairs of sets.

- a) $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$
- b) $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$
- c) $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{1, 2, 3, 4, 5, 6\}$
- d) $A = \{1\}$, $B = \{1, 2, 3, 4, 5, 6\}$

72. Prove that each of the properties in parts (a)–(d) holds whenever A and B are finite sets.

- a) $J(A, A) = 1$ and $d_J(A, A) = 0$
- b) $J(A, B) = J(B, A)$ and $d_J(A, B) = d_J(B, A)$
- c) $J(A, B) = 1$ and $d_J(A, B) = 0$ if and only if $A = B$
- d) $0 \leq J(A, B) \leq 1$ and $0 \leq d_J(A, B) \leq 1$

****e)** Show that if A , B , and C are sets, then $d_J(A, C) \leq d_J(A, B) + d_J(B, C)$. (This inequality is known as the **triangle inequality** and together with parts (a), (b), and (c) implies that d_J is a **metric**.)

Fuzzy sets are used in artificial intelligence. Each element in the universal set U has a **degree of membership**, which is a real number between 0 and 1 (including 0 and 1), in a fuzzy set S . The fuzzy set S is denoted by listing the elements with their degrees of membership (elements with 0 degree of membership are not listed). For instance, we write $\{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.1 \text{ Oscar}, 0.5 \text{ Rita}\}$ for the set F (of famous people) to indicate that Alice has a 0.6 degree of membership in F , Brian has a 0.9 degree of membership in F , Fred has a 0.4 degree of membership in F , Oscar has a 0.1 degree of membership in F , and Rita has a 0.5 degree of membership in F (so that Brian is the most famous and Oscar is the least famous of these people). Also suppose that R is the set of rich people with $R = \{0.4 \text{ Alice}, 0.8 \text{ Brian}, 0.2 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$.

73. The **complement** of a fuzzy set S is the set $\bar{S}$, with the degree of the membership of an element in $\bar{S}$ equal to 1 minus the degree of membership of this element in S . Find $\bar{F}$ (the fuzzy set of people who are not famous) and $\bar{R}$ (the fuzzy set of people who are not rich).

74. The **union** of two fuzzy sets S and T is the fuzzy set $S \cup T$, where the degree of membership of an element in $S \cup T$ is the maximum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cup R$ of rich or famous people.

75. The **intersection** of two fuzzy sets S and T is the fuzzy set $S \cap T$, where the degree of membership of an element in $S \cap T$ is the minimum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cap R$ of rich and famous people.

Links